

BANARAS HINDU UNIVERSITY

B.A./B.Sc. (Hons.) Semester V Examination, 2016-17

Paper No. STA/STB-502: Statistical Inference and Decision Theory

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer any **five** questions. All questions carry equal marks. Symbols have their usual meanings.*

Question 1.

- (a) Define consistency of an estimator. Let $X_1, X_2, \dots, X_n$ be a random sample from the probability density function:

$$f(x, \theta) = e^{-(x-\theta)}, \quad x \geq \theta, \quad \theta > 0.$$

Verify whether $T_n = X_{(1)}$ (the first-order statistic) is consistent for θ . [7]

- (b) If T_1 is the minimum variance unbiased estimator (MVUE) of $\psi(\theta)$, and T_2 is any other unbiased estimator of $\psi(\theta)$ with efficiency $e = e_0$, then show that the correlation coefficient between T_1 and T_2 is given by:

$$\rho = \sqrt{e_0}.$$

[7]

Question 2.

- (a) Define a sufficient statistic. Let $X_1, X_2, \dots, X_n$ be a random sample of size n from a Poisson distribution with parameter θ . Verify whether (i) $X_1 + X_2$ and (ii) $X_1 + 2X_2$ are sufficient statistics for θ . [7]
- (b) Let $X_1, X_2, \dots, X_n$ be a random sample of size n from $U(0, \theta)$. Find the minimum variance unbiased estimator of θ . [7]

Question 3.

- (a) State and prove the Rao-Blackwell theorem. [7]
- (b) What do you mean by a minimum variance bound estimator? Let $X_1, X_2, \dots, X_n$ be a random sample from:

$$f(x, \theta) = \theta e^{-\theta x}, \quad x \geq 0, \quad \theta > 0.$$

Obtain the minimum variance bound estimator of θ . [7]

Question 4.

- (a) Explain the method of maximum likelihood estimation. Obtain the MLE of θ based on a random sample of size n from the following distributions:

(i) $f(x, \theta) = \theta x^{\theta-1}, \quad 0 < x < 1, \quad \theta > 0$

(ii) $f(x, \theta) = \theta(1-x)^{\theta-1}, \quad 0 < x < 1, \quad \theta > 0$

[7]

- (b) Explain the method of moments for estimating parameters. Let $X_1, X_2, \dots, X_n$ be a random sample of size n from a Binomial distribution $B(n, p)$ with unknown parameters n and p . Obtain the estimators of the parameters by the method of moments. [7]

Question 5.

- (a) What is a confidence interval? What is meant by confidence coefficient? Explain the method of construction of confidence intervals. [7]

- (b) Obtain the $100(1 - \alpha)\%$ confidence interval for σ^2 of the normal distribution $N(\mu, \sigma^2)$ when:

(i) μ is known,

(ii) μ is unknown.

[7]

Question 6.

- (a) Explain the following terms:

(i) Simple and composite hypothesis

(ii) Most powerful test

(iii) Size of a test and level of significance

[7]

- (b) State the Neyman–Pearson lemma. Obtain the most powerful test for testing $H_0: \theta = \theta_0$ against $H_1: \theta = \theta_1$ ($\theta_1 > \theta_0$) on the basis of a sample of size n from an exponential distribution with pdf:

$$f(x, \theta) = \theta e^{-\theta x}, \quad x > 0, \quad \theta > 0.$$

[7]

Question 7.

Describe the procedure of the likelihood ratio test. Obtain the likelihood ratio test for testing $H_0: \sigma^2 = \sigma_0^2$ against $H_1: \sigma^2 \neq \sigma_0^2$ on the basis of a random sample of size n from $N(\mu, \sigma^2)$ with unknown mean μ . [14]

Question 8.

Explain the loss function and the risk function. Let $X_1, X_2, \dots, X_n$ be a random sample from $N(\mu, 1)$. Under the squared error loss function, obtain the Bayes estimator of μ when the prior distribution for μ is $N(\mu_0, \tau^2)$. [14]